

Liu, Hong-Wen

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Summary — Enthusiastic undergraduate student passionate about research in structure formation and evolution, cosmology, and dark matter, with experience in numerical simulations and computational astrophysics.

Education

National Tsing Hua University (NTHU)

Bachelor of Science in Physics

Sep 2022 – Present

Course

- Nonlinear Dynamics and Chaos
- Experimental Physics
- Introduction to Relativity
- Fundamentals of Observational Astronomy
- Quantum Physics
- Thermal and Statistical Physics
- Modern Physics Lab.
- Introduction to Modern Cosmology

Academic Experience

Institute of Astronomy, NTHU

Undergraduate student

Oct 2024 – Present

Supervisors: Prof. Hsiang-Yi Karen Yang

- Learning to **Cool-Core Destruction in Merging Clusters with AGN Feedback and Radiative Cooling**
- Use different parameters (AGNs Feedback, angular momentum, etc.) to research the effect on the merging clusters by using **Hydrodynamic code GAMER-2**

Institute of Astronomy & Astrophysics, Academia Sinica

Undergraduate student

Jul 2025 – Present

Supervisors: Dr. Shouvik Roy Choudhury

- Learning to **Neutrino mass bounds from cosmology**
- Use new parameter neutrino mass to research the effect on the Λ CDM model by using **Cobaya: Code for Bayesian Analysis of hierarchical physical models**
- Investigated the interplay between dark energy and neutrino physics using DESI, Planck, and Type Ia supernovae cosmological datasets.

Teaching Appointments

Department of Physics, NTHU

Teaching Assistant, General Physics Lab

Feb 2025 – Jun 2025

Supervisors: Prof. Jow-Tsong Shy

- Led lab sessions and answered questions on experimental design and data interpretation.
- Graded reports and held office hours for lab-related inquiries.

Department of Physics, NTHU

Teaching Assistant – Thermostatistical Physics

Sep 2025 – Dec 2025

Supervisors: Prof. Kuo-An Wu

- Conducted weekly TA sessions outside of class to reinforce core concepts and problem-solving skills.
- Supported grading and clarified course content in supplementary review sessions.
- Deepened subject mastery through teaching and communicating advanced Thermostatistical mechanics topics.

Community engagement

NTHU Astronomy Club Astrophysics Study Group (AstroRead)

Sep 2022 – Present

A student-led reading group focused on exploring and discussing advanced topics in astronomy and astrophysics.

- Role: member (2022 - 2023), host (2024 - Present)
- Lectures: Dark Matter (2023), Introduction of Astronomy and Astrophysics (2024), Inflation (2025)

NCTS-TCA Summer Student Program 2024 Workshop

Jul 01 2024 – Jul 03 2024

- Completed a 3-day intensive workshop hosted by National Center of Theoretical Sciences
- Courses attended: cosmology, galaxies, etc
- Acquired comprehensive insights into theoretical and computational astrophysics

2024 NTHU Starkids camp

Jul 2024 – Aug 2024

A science outreach program aimed at promoting astronomy education through lectures, hands-on activities, and night-sky observations.

- Role: Director
- Organized six-day-education camp with over 60 partners and attracted over 90 participants.

- Cooperated with Astronomy Club of University Union in Taiwan (ACUUT) to promote education in camp activities.

NTHU Astronomy Club

Hsinchu, Taiwan

The club is dedicated to promoting, popularizing astronomy and providing astronomical educational services

2024 - 2025

- Role: member (2022 - 2023), vice director (2024 - Present)
- Organized over 4 observation camps with 13 partners and each events attracted over 30 participants.
- Organized and conducted observational activities , high-altitude locations (above 2000 meters), guiding participants in telescope operation and star identification.

Skills

Programming language: Python, Bash, $\text{\LaTeX}$, MATLAB, C/C++